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TSS-1R electron currents: Magnetic limited collection from a heated presheath

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Abstract. We show that currents collected by TSS-1R are consistent with Parker-Murphy magnetic limited collection from the ionosphere where the local electron temperatures are elevated as observed during the first TSS-1 mission. An analytic development of the previous Katz et al. [1994] presheath model is presented. This model accounts for the heating that occurs as electrons fall through the potentials necessary to reflect the ram ions while satisfying the Bohm sheath stability criterion. This presheath model is coupled to a sheath sized to the Parker-Murphy collection radius to predict the current collected by TSS-1R. The resultant currents are 2 to 3 times larger than for the standard Parker-Murphy collection model and are consistent with observations.

Introduction

Tether currents measured during TSS-1R [Wright et al., 1996; Thompson et al., 1997] are reported to be factors of 2 to 3 greater than the predictions of the conventional Parker-Murphy ($P-M$) theory [Parker and Murphy, 1967]. This result is surprising since $P-M$ theory provides an upper limit for current collection. It is well known, however, that conventional sheath models are not complete without treatment of the presheath which matches sheath edge conditions to the ambient plasma [Parrot et al., 1982], and that the presheath can provide significant current enhancement. $P-M$ theory computes the limiting radius from which a probe can collect an electron across the magnetic field, and assumes the full electron thermal current is available within that radius or flux tube. This amounts to a 'standard $P-M$' presheath with a less than unity net enhancement since a probe collects over its cross section rather than its surface area.

Although we will develop a presheath model with greater enhancement than the standard $P-M$ presheath, we observe that the plasma's apparent ability to support even the standard $P-M$ presheath is significant. On the scale of the TSS, the surrounding plasma is collisionless and the ambient electron gyroradius is small. So it would seem that the TSS should deplete the immediate flux tube of electrons. If the tube was refilled along a length on which the plasma can be considered collisional [Sanmartin, 1970], it is unclear whether the tube could carry a full thermal current to the sheath surface as assumed by $P-M$ theory.

There are experimental data which suggest that the flux tubes indeed carry the full electron thermal flux to the sheath edge. Although many electron beam emitting rockets have observed return currents in excess of the $P-M$ limit, it was demonstrated by the CHARGE-2 experiment that this effect was due to the beam [Myers et al., 1989]. Here, the electron emitting mother payload experienced a return current in excess of the $P-M$ limit, while the tethered daughter did not [Mandell et al., 1990]. The SPEAR I experiment also observed electron collection consistent with magnetic limiting in the high voltage sheath of a positively biased probe [Katz et al., 1989]. Both the NASCAP/LEO and POLAR codes used in these studies [Mandell et al., 1990; Katz et al., 1989] employ steady state electron tracking in space charge sheaths that in the case of POLAR are computed self consistently [Lilley et al., 1989]. These codes have been shown to produce $P-M$ limiting for spherical probes [Katz et al., 1989]. Significantly, the agreement between these codes and experiments is obtained with the assumption of sheaths fed by flux tubes carrying a full electron thermal current.

These sounding rocket experiments demonstrate that for even near stationary collection, the ionosphere can refill the flux tube near the sheath. Understanding how this refilling occurs is a important part of magnetic probe theory [Laframboise and Sonmor, 1993], and the ultimate source of the electrons constitutes the current closure issue [Stenzel and Urrutia, 1997]. We argue that separating the issues of current closure and current collection is empirically justified. This allows us to construct a 'local' probe model where we can assume an undisturbed plasma exists on the boundary of the interaction.

Positive Probe Sheath and Current

The disturbed region of plasma surrounding a high voltage probe is commonly dissected into sheath and presheath regions. In the sheath, the positive potential repels ions and the space charge is dominated by electrons. The sheath edge is usually considered to be thin. The presheath is a quasi-neutral region that matches the sheath to the undisturbed plasma.

There are three basic sheath models, 'Orbit Limited', 'Space Charge Limited', and 'Magnetic Limited'. If the probe radius, R_o, is comparable to or smaller than the ambient Debye length, $\lambda_D = \sqrt{\varepsilon_o kT/Ne^2}$, the (thermal) angular momentum of the collected particles is significant. This is called 'Orbit Limited' collection [Laframboise and Parker, 1973], and the current varies linearly with the potential. The more relevant regime here is $R_o \gg \lambda_D$, where a generally inviolate limit was presented by Langmuir and Blodgett [1924] for the case of no magnetic field and negligible ambient electron energy compared to the probe potential. For an infinitesimally thin sheath, the spherical model of Langmuir and Blodgett reduces to a Child-Langmuir 1D space charge limited diode [Child, 1911; Langmuir, 1913], which is used here to introduce a practical parameterization. This expres-

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sion for the current density, J, crossing a sheath gap, D, with potential drop, V, is

$$J = \frac{4}{9}\varepsilon_o\left[\frac{2e}{m_e}\right]^{1/2}\frac{V^{3/2}}{D^2} \tag{1}$$

Setting this equal to the thermal current, $J_{th} = Ne\sqrt{kT/2\pi m}$,

$$D = 1.26\lambda_D\left[\frac{eV}{kT}\right]^{3/4} \tag{2}$$

For a spherical sheath of finite thickness, we adapt an approximation from Parker [1980]. With sheath radius R_{LB}, and conditions such that $R_0/\lambda_D \gtrsim 10$, and $D/R_0 \gtrsim 10$, we have;

$$\frac{R_{LB}}{R_0} \approx \left[\frac{D}{R_0}\right]^{\alpha} \quad 1/2 < \alpha < 4/7 \tag{3}$$

The total current, I, is J_{th} summed over the sheath surface.

$$I \approx 4\pi J_{th} R_0 \lambda_D\left[\frac{eV}{kT}\right]^{\beta} \quad 3/4 < \beta < 6/7 \tag{4}$$

The space charge limit expresses the principle that the shielding charge fundamentally limits the extent of the probe potential and thus the current a probe can collect. It is thus a strong limit.

Another limit is given by the theory of Parker and Murphy [1967]. They observed that for a potential distribution symmetric about the magnetic field, the conservation of canonical angular momentum and energy yields a maximum cylinder radius, R_{PM}, from which an electron may reach the probe surface.

$$R_{PM} = R_o\left[1+\left[\frac{8eV}{m\omega_{ce}^2R_o^2}\right]^{1/2}\right]^{1/2} \tag{5}$$

This radius defines a flux tube from which electrons may be collected and leads to a limiting current after specifying the current density in the flux tube as discussed previously. The $P-M$ limit is weak because it can be violated by fluctuations [Linson, 1969] or lack of symmetry [Katz et al., 1989].

Whether the probe current is $P-M$ or space charge limited can be estimated by forming the ratio,

$$\frac{R_{LB}}{R_{PM}} \approx \left[\frac{\omega_{ce}}{\omega_{pe}}\right]^{1/2}\left[\frac{eV}{kT}\right]^{1/8} \tag{6}$$

Whenever $R_{LB}/R_{PM} > 1$, the collection is $P-M$ limited, and the space charge of electrons within R_{PM} does not completely shield the probe. The probe potential therefore extends beyond R_{PM}, creating a region to which electrons are attracted but from which they cannot be collected. Even though these electrons are energetically capable of escaping, they cannot easily find an exit vector and form an $\mathbf{E}\times\mathbf{B}$ drifting shell of quasi-trapped electrons. Self-consistent modeling [D. Cooke, unpublished] and laboratory experiments [Antoniades et al., 1990] in support of the SPEAR I experiment demonstrate that this layer of circulating charge mostly overlies the $P-M$ sheath region with negligible extent beyond R_{PM}. This contributes additional space charge and confines the sheath close to the $P-M$ radius over a wide range of $\omega_{ce}/\omega_{pe} > 0.5$. Figure 1 illustrates this region. The actual trajectories of the quasi-trapped electrons reach from this shell to just above the probe surface.

A Heated Presheath Model for the TSS

The presheath [Parrot et al., 1982] matches conditions at the sheath edge to the ambient plasma. At the edge of a high (positive) voltage sheath, the ion density transitions rapidly from a value near zero inside the sheath to near ambient just outside. Since the sheath is mostly absorbing there are no outgoing electrons. Were it not for the presheath, the sheath edge electron density would be only about half of ambient. The role of the presheath is to concentrate the attracted species so that the Bohm sheath criterion [Bohm, 1949; Riemann, 1991] is satisfied everywhere. The Bohm criterion requires that throughout the shielding region surrounding a probe the sign of the charge density should be opposite that of the probe. Any serious deviation from this condition leads to the unphysical picture of a probe shielded by like charge.

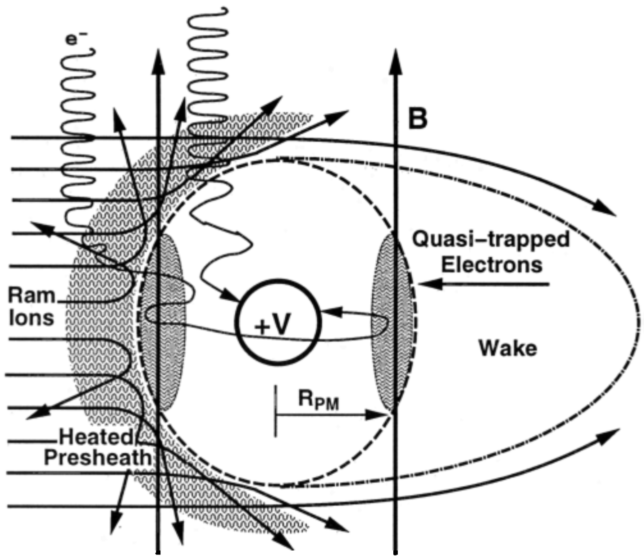

Figure 1. Schematic illustration of the proposed interaction model. Reflected ram ions create a heated unmagnetized presheath while the sheath is limited to the Parker-Murphy [1967] radius by quasi-trapped electrons.

Figure 1 illustrates our model. The positive subsatellite repels the ram ions creating an upstream ion density enhancement that must be matched by the electrons for a Bohm stable sheath. This enhancement has been calculated by Katz et al. [1994], Singh and Leung [1997], Ma and Schunk [1997], and Laframboise [1997]. Although these estimates of the peak enhancement vary from 3% to 30%, we observe it to be negligible over most of the region where the potential is elevated, so an approximation of unity throughout the presheath is suitable for the present treatment. Katz et al. [1994] demonstrated that magnetically constrained electrons cannot match even a constant ambient ion density in the presheath, and it was postulated that this Bohm unstable sheath suffers fluctuations sufficient to establish an unmagnetized electron fluid. The original model of Katz et al. [1994] required numerical solution. A fully analytic solution has since been found and is presented here. The equations to be solved are:

$$\text{Electron current, } \mathbf{J}: \ \frac{\mathbf{J}}{\sigma} = \mathbf{J}' = \nabla V - \frac{1}{n}\nabla(nT') \tag{7}$$

$$\text{Heat flux, } \mathbf{Q}: \ \frac{\mathbf{Q}}{\sigma} = \mathbf{Q}' = -\frac{3}{2}T'\nabla T' + \frac{5}{2}T'\mathbf{J}' \tag{8}$$

$$\text{Current continuity: } \nabla\cdot\mathbf{J}' = 0 \tag{9}$$

$$\text{The heat equation: } \nabla\cdot\mathbf{Q}' = \mathbf{J}'^2 \tag{10}$$

Here, V is the electric potential, $T' = kT/e$, T is the electron temperature, k is Boltzmann's constant, and $\sigma = \varepsilon_o\omega_{pe}^2/\nu$ is the electrical conductivity due to the scattering rate, ν, induced by the instability. Notice that the scattering rate does not appear in the final equations, so it appears that ν need only be sufficient for thermal equilibrium. If we assume a constant ion density as discussed above, we may integrate these equations and derive jump conditions for the presheath. To simplify the notation we now drop all primes. Since $\nabla n = 0$, the current equation becomes,

$$\mathbf{J} = \nabla V - \nabla T \tag{11}$$

The heat equation (10) may be expanded and reduced to a quadratic expression if we assume $\nabla^2 V$ to be negligible. Presheath quasi-neutrality suggests that this may be reasonable, and in most cases the charge density ($\varepsilon_o \nabla^2 V$) in the presheath will be small compared to that in the sheath. Rigor however, requires this term to be negligible compared to the remaining terms in equation (12) below, so we identify this as a critical assumption and proceed. The heat equation becomes,

$$\nabla \cdot \mathbf{Q} = -\frac{3}{2}(\nabla T)^2 + \frac{5}{2}\mathbf{J}\nabla T = \mathbf{J}^2 \tag{12}$$

We now substitute for the current and obtain a quadratic expression for δT across the presheath due to the presheath potential drop, δV. Choosing the lower temperature solution yields,

$$\delta T = 2/5\,\delta V \tag{13}$$

Inside the sheath, we expect the accelerated electrons to be controlled by the magnetic field and the sheath radius to be given by the $P-M$ relation. The space outside of the sheath may be divided into two hemispherical regions. In the wake, the dearth of ions leads to a space charge barrier impeding electron collection so it is only the ram hemisphere which provides electrons to the sheath. On the ram side, we apply our presheath model locally to the sheath surface where the sheath edge potential, V_S, is that necessary to repel the normal component of the ram ion velocity. This is $V_S = E_i \cos^2\theta$, where E_i is the ram ion energy, and θ is the angle between the sheath surface normal and the ram direction. We also have $\delta V_S = V_S$ since the presheath begins at zero potential. For $\theta = 0$ the peak enhancement is, $\delta T_P = 2/5\,E_i$, and the electron temperature on the ram side sheath is,

$$T_S(\theta) = T_o + \delta T_P \cos^2\theta \tag{14}$$

The heated presheath thermal current J_S at the sheath surface is,

$$J_S(V_S) = eN_e\sqrt{kT_S(V_S)/2\pi m_e} = J_o\sqrt{T_S(V_S)/T_o} \tag{15}$$

The presheath current, I_{PS}, to the subsatellite as functions of the satellite potential, V_{Sat}, and B, is found by integrating J_S over the ram hemisphere of a spherical sheath with radius, R_{PM}.

$$I_{PS} = 2\pi R_{PM}^2(V_{Sat})\,J_o\,T_o^{-1/2}\int_0^{\pi/2} T_S^{1/2}\sin\theta\,d\theta \tag{16}$$

Defining $\chi = \delta T_P / T_o$ we find,

$$\begin{aligned} I_{PS} &= 2\pi R_{PM}^2 J_o \frac{1}{2}\left[(1+\chi)^{1/2} + \chi^{-1/2}\ln(\chi^{1/2} + (1+\chi)^{1/2})\right] \\ &\approx 2\pi R_{PM}^2 J_o \frac{1}{2}(1+\chi)^{1/2} \end{aligned} \tag{17}$$

The total current, I, is I_{PS} plus the half of the nominal $P-M$ current not included in the ram hemisphere. Thus we have,

$$I \approx 2\pi R_{PM}^2 J_o \frac{1}{2}\left[1 + (1+\chi)^{1/2}\right] \tag{18}$$

and our predicted current enhancement is $\frac{1}{2}(1 + (1+\chi)^{1/2})$.

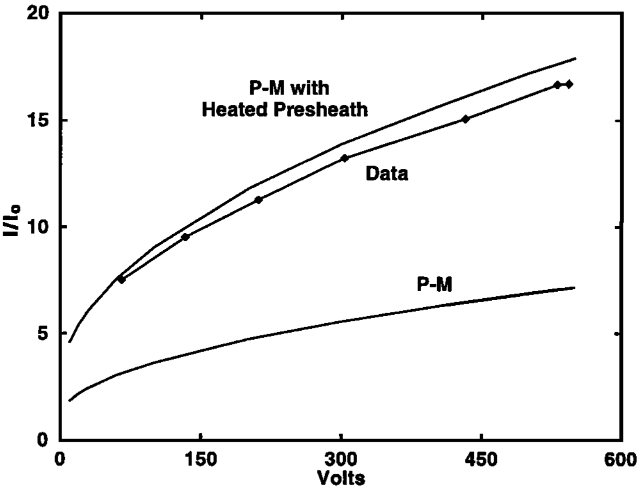

Figure 2. TSS1-R Current-Voltage relationship [Wright et.al. 1996] showing the observed data along with the nominal Parker-Murphy ($P-M$) current and $P-M$ current enhanced by the presheath heating model.

Comparison With Published Data

Wright et al. [1996] published an I-V curve for a single sweep performed at 057/01:08:25. Ambient plasma conditions were: $N_e = 7\,x\,10^{11}\,m^3$, $T_e = 1500\,K = 0.13\,eV$, $B = 0.32\,Gauss$. The satellite radius was $R_o = 0.8\,m$. For $E_i = 4.8eV$, our model gives a peak temperature enhancement of $\delta T_p = 1.9\,eV$, and a current enhancement of 2.5. Figure 2 is adapted from Wright et al. [1996] and shows the reported data, the $P-M$ current, and our enhanced $P-M$ current. For these data the enhancement over $P-M$ is about 2.4 compared to our value of 2.5. This enhancement was observed to be nearly constant for all potentials above 5 or 10 volts. This is also the case for the total TSS-1R data set presented by Thompson et al. [1997] where the enhancements over $P-M$ varied from 2.2 to 2.9. The constancy of the enhancement is as significant as the actual value. It confirms that for higher potentials, the enhancement is independent of voltage, which is consistent with our model.

Discussion

It is instructive at this point to review the physical elements and assumptions of our model. We are proposing three distinct regions: a charge neutral presheath, where we apply the Bohm criterion, and assume thermodynamic equilibrium and unmagnetized flow of electrons; a sheath region where the electron motion is described by Parker and Murphy [1967]; and a transition region with quasi-trapped electrons providing the space charge necessary to confine the sheath within the $P-M$ radius. Although our model is constructed heuristically with knowledge of the observations, two of the three elements are standard in probe theory and there are no free parameters. The quasi-trapped electron region is somewhat new and can be viewed as a critical assumption along with assumption of negligible space charge in the presheath.

Our presheath is assumed to be in thermodynamic equilibrium. Although heating is a recognized mechanism for achieving Bohm stability [Riemann, 1991], full thermodynamic equilibrium is not required. Laframboise [1997] constructed an alternative presheath assuming isothermal electron collection and achieved even greater current concentration. Here though, we appeal to the agreement between our model and the TSS-1R observations as validation of our approach. Further support is provided by the particle simulation of Singh and Leung [1997]. If their simulation region is sufficiently large for the boundaries to not affect the results, the presheath must be included. They report electron energization several radii away from the body in the ram direction where the ion density is greater than or equal to ambient.

Another key element is the unmagnetized presheath electrons. Although the magnetic field may play a role in the details of scattering and heating, our treatment would be unchanged if no magnetic field were present. Without reference to our model, it can be argued that cross field transport of electrons is mandated by the observations. Strict magnetization does not yield even the $P-M$ limit unless some transport fills the flux tubes with the ambient thermal current. We assert that amplification of the field aligned current by factors of 2 to 3 over ambient at large distance from the subsatellite is unreasonable. Such currents are unstable to double layer formation and subsequent restriction to the thermal current [Carlqvist, 1972]. Local cross field transport appears to be the only possible mechanism for concentrating the electron current. In light of the previous argument, the overall $P-M$ characteristic of the observations is all the more intriguing. We suggest that the presheath is a plasma with two components that can participate in collective interactions. The sheath however, contains only electrons which are not resident long enough to participate in any instability. Thus, the sheath is sufficiently stable to support the constants of the motion. We note however, that the quasi-trapped electrons do have long residence times and have been modeled as participating in collective oscillation with the probe charge [Mandell and Katz, 1993].

In summary, we have presented a simple model of electron collection by a large positive sphere moving through the ionosphere. The model is grounded in established theories of plasma probes and observations from the first flight of TSS-1 which showed heating of electrons in the ionosphere near the satellite. The predictions of our model are in remarkable agreement with the published TSS-1R data. As was true when the original paper [Katz et al., 1994] was published, the detailed mechanism that generates the scattering and the scattering frequency is still not understood.

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